Desirable Properties of OLS Estimators-the Gauss-Markov Theorem

1. Introduction

Ordinary Least Squares (OLS) is a widely used method in econometrics for estimating the parameters (α *and* β) of a linear regression model. The Gauss-Markov Theorem establishes the desirability of OLS estimators under certain assumptions, proving that they are the Best Linear Unbiased Estimators (BLUE). Following are the desirable properties of OLS estimators:

2. Desirable Properties of OLS Estimators

For an estimator to be considered desirable, it should possess the following properties:

a) Linearity

An estimator is linear if it is a linear function of the dependent variable. The OLS estimators are derived as linear combinations of the observed values of the dependent variable.

b) Unbiasedness

An estimator is unbiased if its expected value equals the true population parameter: This property ensures that, on average, the OLS estimator correctly estimates the population parameter.

c) Efficiency (Minimum Variance)

An estimator is efficient if it has the smallest variance among all unbiased estimators. OLS estimators achieve minimum variance among the class of linear unbiased estimators under the assumptions of the classical linear regression model.

d) Consistency

An estimator is consistent if it converges in probability to the true parameter value as the sample size increases. OLS estimators are consistent if the assumptions of the classical regression model hold.

e) Sufficiency

An estimator is sufficient if it captures all available information about the population parameter from the sample data. OLS estimators are sufficient under normality assumptions.

f) Asymptotic Normality

As the sample size increases, OLS estimators tend to follow a normal distribution due to the Central Limit Theorem (CLT), making statistical inference feasible.

The **Gauss-Markov Theorem** states that under the classical linear regression model assumptions, the OLS estimator is the **Best Linear Unbiased Estimator (BLUE)**. This means it has the smallest variance among all linear and unbiased estimators.